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SEIZURE DISORDERS

Seizures :

Definition : Abnormal paroxysmal electrical activity in brain.

Manifestations :

- motor (tonic-clonic seizures).
- Sensory (tingling, numbness, abnormal perceptions).
- Behavioural changes.
- Autonomic disturbances : Sudden variation in heart or respiratory rate.

Causes : $\left\{ \begin{array}{l} \text{Provoked : Eg. Fever in febrile seizures.} \\ \text{Unprovoked : underlying brain injury/dysfunction.} \end{array} \right.$

Epilepsy :

- Recurrent (≥ 2 episodes) seizures.

Febrile Seizures

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Definition :

Seizures provoked by fever alone without :

- underlying CNS infection or,
- Previous unprovoked seizure.

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Incidence :

- m/c cause of seizure in childhood.
- Age affected : 6 months to 5 yrs.

Developmental status of patients : Normal.

Types :

	Typical/ Simple	Atypical/ Complex/ Complicated
Presentation	General tonic-clonic seizures (GTCS)	Focal onset
Incidence	more common	Less common
Episodes of seizures	1 during febrile illness	multiple seizures during the same febrile illness
Duration	< 15 min	> 15 min

RECURRENCE

- $\uparrow$ risk of recurrence upto 5 yrs of age.

Risk factors until 5 yrs of age :

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major :

- Age < 1 yr.
- Duration of fever < 24 hr.
- Fever > 38 to 39°C (100.4 to 102.2°F).

minor :

- Family h/o febrile seizures/epilepsy.
- Complex febrile seizures.
- male gender.
- Hyponatremia at presentation.
- Child in day care settings.

No. of risk factors	Risk of recurrence
0	12 %
1	25 - 30 %
2	50 - 59 %
≥ 3	73 - 100 %

Risk factors for epilepsy > 5 yrs of age :

Risk factors	Risk for subsequent epilepsy
Simple febrile seizure	1%
Recurrent febrile seizure	4%
Complex febrile seizure (>15 min in duration or recurrent within 24 hr)	6%
Fever <1 hr before febrile seizure	11%
Family history of epilepsy	18%
Complex febrile seizure (Focal)	29%
Neurodevelopmental abnormalities	33%

Low-risk factors

High risk factors →

Investigations :

- EEG.
- MRI.

LUMBAR PUNCTURE :

Goal : To rule out CNS infections (meningitis).

Indications :

- Any signs of CNS infection (meningeal signs).
- Recent antibiotic course (Can mask sign of meningitis).
- unimmunized / unknown vaccination status.

MANAGEMENT :**Treatment :**

Control of fever :

Antipyretics (Eg : Paracetamol).

Tepid sponging : Cotton dipped in cold water.

a. Control of epilepsy :

DOC : IV Benzodiazepines

(Diazepam, lorazepam, midazolam)

↓ If no IV access

Per rectal diazepam/intranasal midazolam

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Intermittent prophylaxis :

Given during first 2 days of fever in home setting.

Indications :

- Major risk factors for recurrence.
- Frequently recurrent febrile seizures :
 - > 3 episodes in 6 months OR,
 - > 4 episodes in 1 year.

DOC : **Oral clobazam** (In India) / Oral diazepam.**Absence Seizures (Petit mal seizures)**

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minimal manifestations.

Age affected : School going children (> 5 yrs).

Clinical features : **Last upto 30 sec.** Precipitating factors :

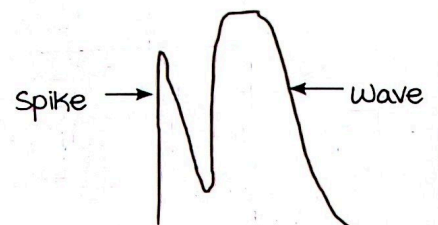
- Vacant stares.
- Eye flutters.
- **Hyperventilation** (Used to diagnose clinically).
- Stress.

Types :

- Typical : minimal manifestations.
 - **3 Hz spike & wave pattern.**
- Atypical : Can have GTCS/ myoclonic seizures.
 - 10 Hz spike & wave pattern.

EEG findings :

- 3 Hz or 3/sec spike & wave in typical absence seizures.



Treatment :

- DOC in India : **Valproate.**
- 2nd line drug : **Lamotrigine.**

Note :

Ethosuximide : Limited availability & use in India.

Juvenile Myoclonic Epilepsy (JME) (JANZ syndrome)

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myoclonus : Sudden, jerky movements d/t sudden contraction of muscle groups.

Age affected : Adolescents.

Inheritance : Polygenic inheritance.

Associated with mutations of :

- **CLCN2**
- **GABRA1**
- **myoclonin**
- **CACNB4**

Clinical features :

- myoclonic episodes : Occurs upon awakening (Eg : Inability to hold toothbrush d/t jerks).
- **Consciousness unaffected.**
- A/w other seizure patterns in some patients : GTCS/ Absence seizures.

Precipitating factors :

- Sleep deprivation.
- Alcohol intake.

EEG findings :

- **4 - 6 Hz polyspike & wave pattern.**

Treatment :

- DOC : **valproate.**
 - Life long treatment : D/t ↑ risk of adulthood seizures.

Epilepsy Syndromes

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Associated with injury to brain (low IQ, developmental abnormalities).

WEST SYNDROME

Onset : < 1 yr.

Types :

- Idiopathic (30 %) :
Not a/w other CNS abnormalities.
- Symptomatic (70 %) : Worst outcome.
CNS defects present.

Clinical features :

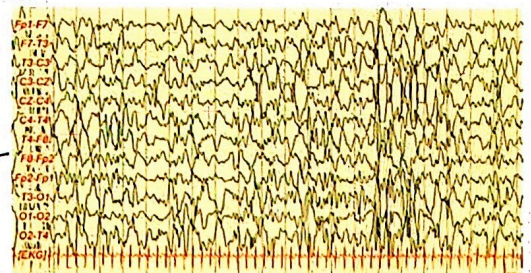
- Infantile spasms :
 - **Jack-knife seizures** (Abrupt contraction of limbs).
 - **Salaam spells** (Repeated neck flexion).

Developmental regression
(Loss of previously
acquired milestones)

Onset < 1 yr



High voltage spikes in
chaotic background



Hypsarrhythmia

Treatment :

- Initiate at the earliest (Delay → Irreversible brain damage).
- DOC : Inj. **ACTH.**
- Second line drug : **Vigabatrin** (DOC if associated with tuberous sclerosis).

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Nervous System

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DRAVET SYNDROME: (severe myoclonic epilepsy of infancy).

Onset: < 1 yr.

Genetic defect:

- SCN1A gene (1 alpha subunit of sodium channel).

Clinical features:

- multiple seizure types seen (Difficult to manage).

management:

- First line drug: valproate (Unsatisfactory treatment response).
- Newer drug: **Stiripentol** → ↑ GABA activity → ↑ seizure.

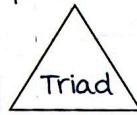
LENNOX GASTAUT SYNDROME:

Onset: 2 to 10 yr.

Clinical features:

Treatment: **Valproate** (DOC).

Developmental delay

multiple refractory
seizuresTypical EEG: 1.5 to 2.5 Hz
spike & wave pattern**Status Epilepticus**

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Emergency characterised by long duration seizure.

Time lines:

- T_i : Time at which treatment should be initiated → > 5 min.
- T_a (Indicates prognosis): Time after which seizure activity may cause long term consequences (Neuronal injury) → > 30 min.

Treatment algorithm :

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Child presents with seizures > 5 min (At T)

Stabilise airway, breathing, circulation & disability (Neurological)

+
Attempt IV access

Blood glucose management

If hypoglycemic (< 54 mg/dl)

If normal

First line anti-epileptic IV benzodiazepines (midazolam/Lorazepam)

Continued Immediate push of 5 ml/
seizures kg of 10 % dextrose

No response

If resolved

monitoring

Second line anti-epileptics :

One of { Fosphenytoin
valproate
Levetiracetam

No response

Refractory status epilepticus :

• Anaesthetic dose of one of :

- Thiopental.
 - midazolam.
 - Phenobarbital.
 - Propofol.
- + }
- Continuous IV infusions

• EEG monitoring for seizure activity.